CODE APEX

THE END IS COMING.

verzii

Team Details

- Team name - Dumb Potato's
- Team leader Name - Balaji Bandgar
- Problem Statement (Domain) - Autonomous AI QA Engineer (AI Based Problem Statements Powered by iMocha)



Problem Statement & Target Need

A typical scenario

- Friday 6 PM: a 3-person startup ships a checkout update.
- A broken coupon field slips through. No tester, no time.
- Customers find the bug first.



What these teams need

- A tester that explores on its own, with no scripts to write
- Bugs proven with screenshots and exact steps
- Tests that are saved and re-run after every release

WHO IS AFFECTED

Small teams and startups shipping fast, with no dedicated tester.

THE GAP

- Manual testing is slow
- Automated tests go stale
- Bug reports lack proof

WHY NOW

Faster releases demand continuous, reliable testing.

TODAY
Manual clicking, vague bug reports, tests that go stale

→

WITH TESTPILOT AI
Agent explores, finds bugs, attaches screenshots and exact steps

AI can explore, find bugs, and prove them.

The cost of waiting

- Lost user trust, weekend hotfixes, and endless “can’t reproduce” threads

Proposed Solution & Innovation

What makes it different

- Autonomously explores web applications using AI-powered browser automation.
- Generates test cases for normal flows, edge cases, and invalid inputs.
- Identifies and verifies bugs with screenshots and reproducible evidence.
- Converts successful user journeys into reusable Playwright tests.

Core mechanism

- EXPLORE: Stagehand discovers pages, forms, buttons, and navigation.
- PLAN & EXECUTE: LLM generates test cases; Stagehand interacts with the browser.
- VERIFY: Playwright assertions distinguish actual bugs from test errors.
- REPORT: Generate bug reports with reproduction steps, screenshots, and reusable tests.

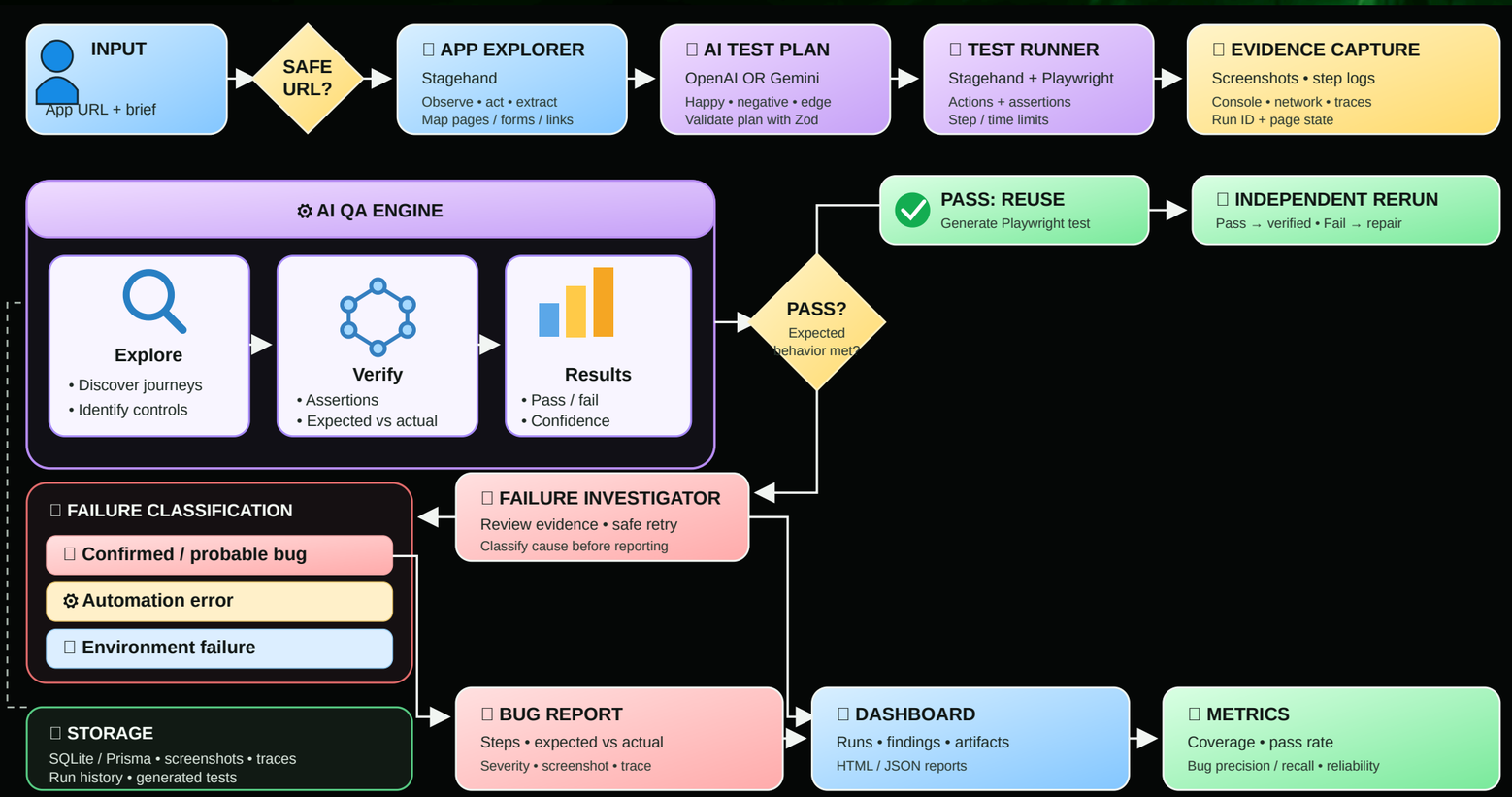
Why it's feasible in this timeframe

- Uses existing LLM APIs instead of training a model.
- Stagehand and Playwright provide browser automation building blocks.
- Built with familiar technologies: React/Next.js, Node.js, and TypeScript.
- A controlled sample application with seeded bugs enables measurable evaluation.

System Architecture & Tech Stack

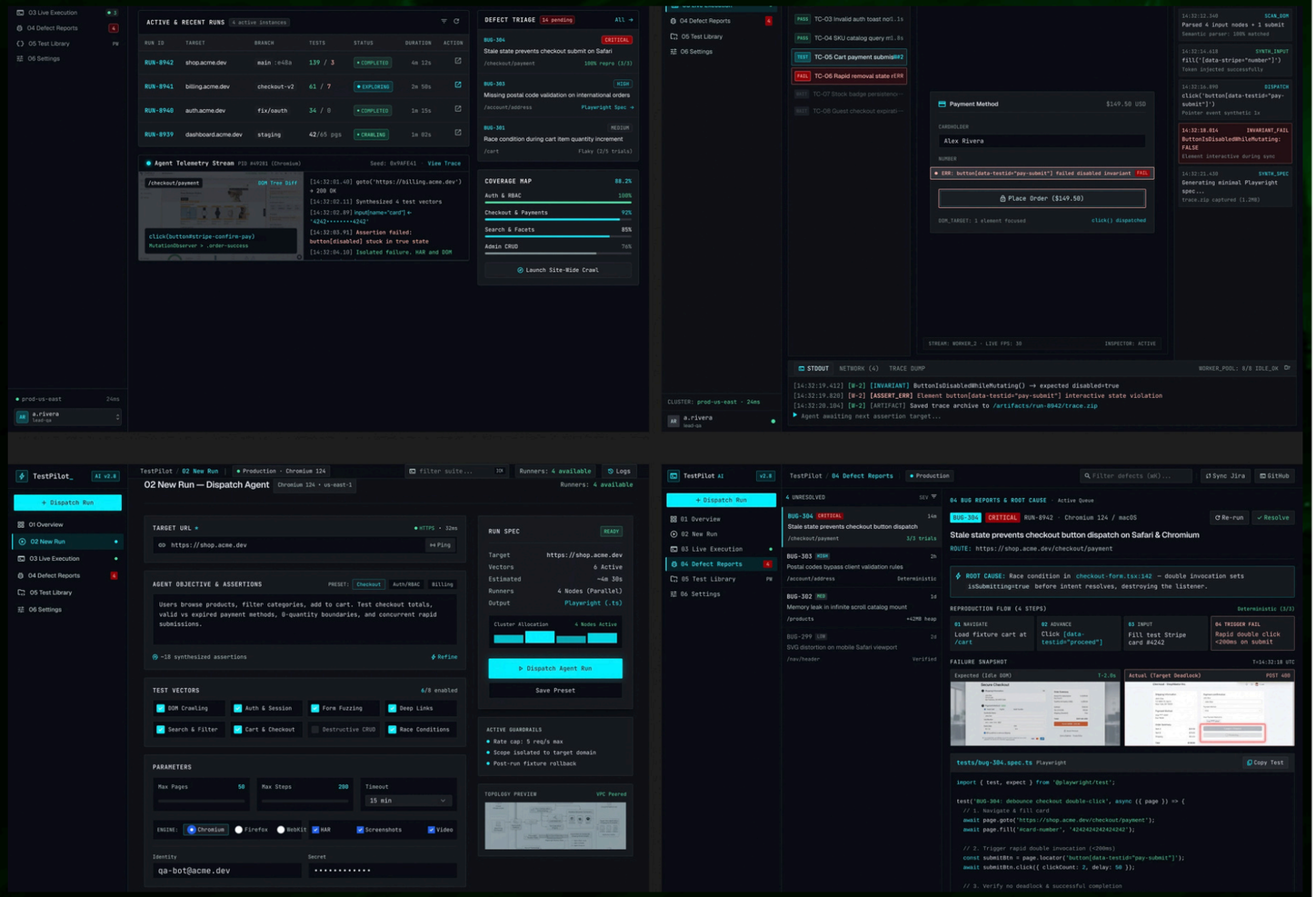
TECH STACK

- **Frontend**
- **Backend**
- **Database**
- **AI / APIs**
- **Hosting / DevOps** — Vercel, Render



Key Features & UI/UX

- Live AI Mission Control: Real-time view of agent actions and testing progress.
- Interactive Browser Replay: Watch and replay the agent's testing steps.
- Evidence-Based Bug Reports: Screenshots, reproduction steps, and expected vs. actual results.
- Bug Verification Center: Distinguishes real bugs from automation errors.
- Application Journey Map: Visualizes tested and unexplored user flows.
- Self-Building Test Library: Converts successful journeys into reusable Playwright tests.
- Regression Radar: Highlights newly introduced and recurring bugs.
- QA Health Dashboard: Displays bug detection, test coverage, and reproducibility metrics.



[Prototype-link]

Implementation Roadmap & Scope

What's built today, what's in progress, and what's planned beyond the hackathon

● BUILT TODAY

Live dashboard prototype
Run setup, live agent view, bug reports

Real browser testing
Stagehand + Playwright on a live journey

Seeded-bug sample app
Planted defects for fair measurement

Evidence capture
Screenshots, console logs, traces

● IN PROGRESS

Failure investigator
Real bug vs test error vs environment

Auto-generated tests
Reusable Playwright tests, re-run to verify

Benchmark results
Bug precision and recall on seeded bugs

Multi-journey runs
Dashboard fed by live run data

● BEYOND HACKATHON

CI/CD integration
Run on every push with GitHub Actions

Regression tracking
New and recurring bugs across builds

Scale out
Multi-app testing, parallel cloud browsers

Issue export
One-click bugs to GitHub and Jira

In scope: web apps, login, search, forms and cart journeys, evidence-backed bug reports, reusable tests

Out of scope for now: payments, destructive actions, mobile apps, load testing

Stage 2: one journey, one bug report, one reusable test. **Stage 3:** multiple journeys, benchmark, full dashboard.

Expected Impact & Scalability

Who benefits, how big the opportunity is, and how this grows beyond a demo.



Who benefits?

Startups, small development teams, QA engineers, and developers shipping web applications.



Immediate impact

Reduce repetitive manual checks, detect defects earlier, and produce reproducible bug reports with evidence.



Growth opportunity

Expand from one user journey to multi-page applications, regression suites, and multiple projects.



Beyond the demo

Integrate with CI/CD pipelines, track regressions across builds, and reuse generated Playwright tests.

Scalability Roadmap

01 MVP
- One app • Core journey
- AI-generated tests
- Evidence-backed bug reports

02 Expand
- Multiple journeys
- Agent roles
- Reusable tests & QA dashboard

03 Scale
- Multi-app testing
- CI/CD integration
- Regression suites & team workflows

Success Metrics

- Bug precision & recall
- Journey coverage
- Test reliability
- Time to detect

VISHWAKARMA INSTITUTES VI

AISF
ARTIFICIAL INTELLIGENCE STUDENT FORUM

VCA

CODE APEX

BUILD BEFORE IT DOES.

POWERED BY verzii

THANK YOU